Comparison of Text Feature Representations and Classification Models for Fake News Detection

Mohamed Omari & Harn Zhin Chong (Group 4)

Ironhack Bootcamp - AI Engineering

24.04.2026

Objective

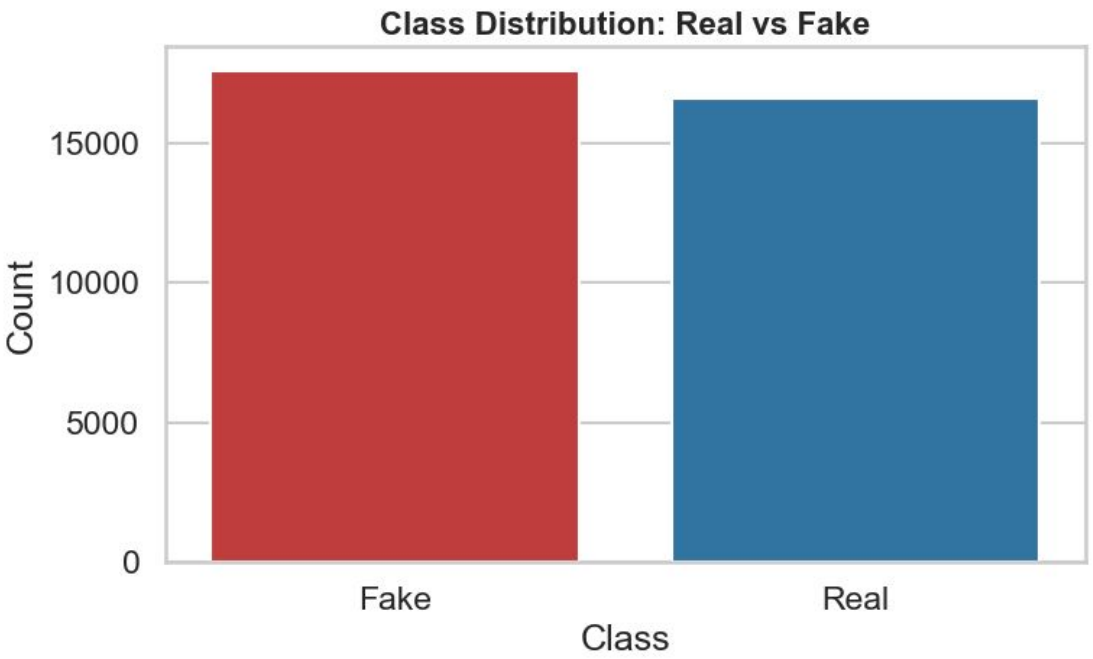
- Build a model to classify news as Real vs Fake.
- Compare different text preprocessing, feature representations, and models.

Hypothesis

- Advanced representations (e.g., Word2Vec, BERT embeddings) will outperform classical methods like BoW or TF-IDF.

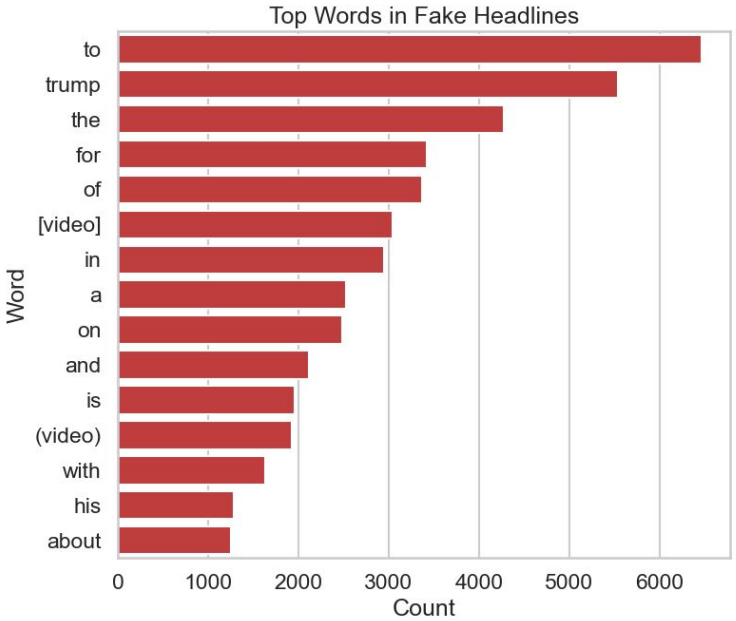
Data Overview

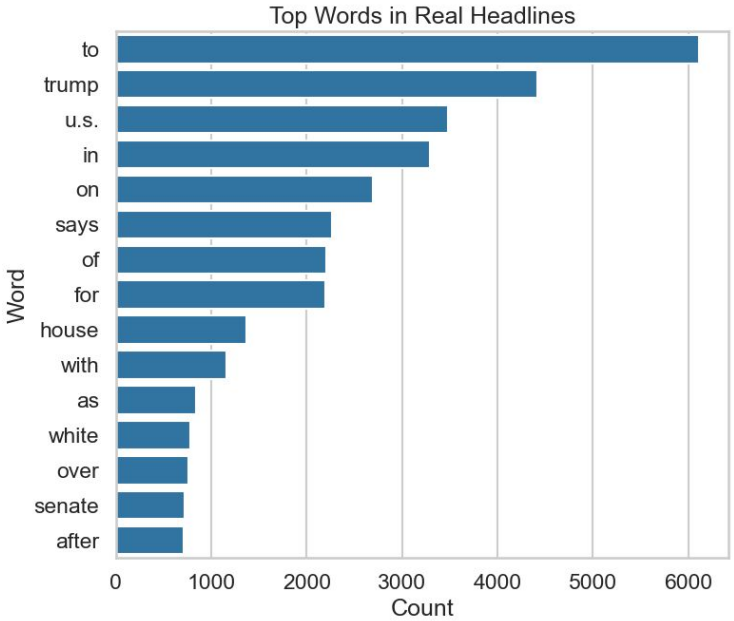
Number of headlines: 34152



Data Overview

- Fake and real headlines share many common top words (e.g. “to”, “Trump”).
- The word “video” is mostly mentioned in the fake headlines.





Fake News Detection – Experimental Workflow

1. Data

News Articles

Labels (Real / Fake)

Train / Validation / Test Split

(Stratified Split)

2. Preprocessing

- split_lemma
- split_stem
- token_lemma
- token_stem

4 Preprocessing Variants

3. Feature Representation

Sparse–Text Representations

- BoW (Bag of Words)
- TF-IDF
- TF-IDF (Log-scaled TF)

Embedding Representations

- Word2Vec (Dense Vectors)
- BERT (Contextual Embeddings)

5 Feature Families (3 Sparse + 2 Embedding)

4. Model Families

Sparse–Text Models (High-dimensional)

- Naive Bayes
- Logistic Regression
- Random Forest
- SVM
- XGBoost
- Neural Network

Embedding Models (Dense Vectors)

- Gaussian Naive Bayes
- Logistic Regression
- Random Forest
- SVM
- XGBoost
- Neural Network

6 Model Families (Per Feature Family)

5. Evaluation

Cross-Validation (on Training Set)

Hyperparameter Tuning

Train Best Model (on Train Set)

Evaluate on Validation Set

6. Results & Comparison

Compare All Pipelines (84 Configurations)

- Train Accuracy
- Validation Accuracy
- CV Accuracy

Analyze:

- Best Performing Models
- Overfitting (Train–Val Gap)
- Effect of Preprocessing
- Effect of Feature Family
- Effect of Model Family

TOTAL EXPERIMENT CONFIGURATIONS: 84

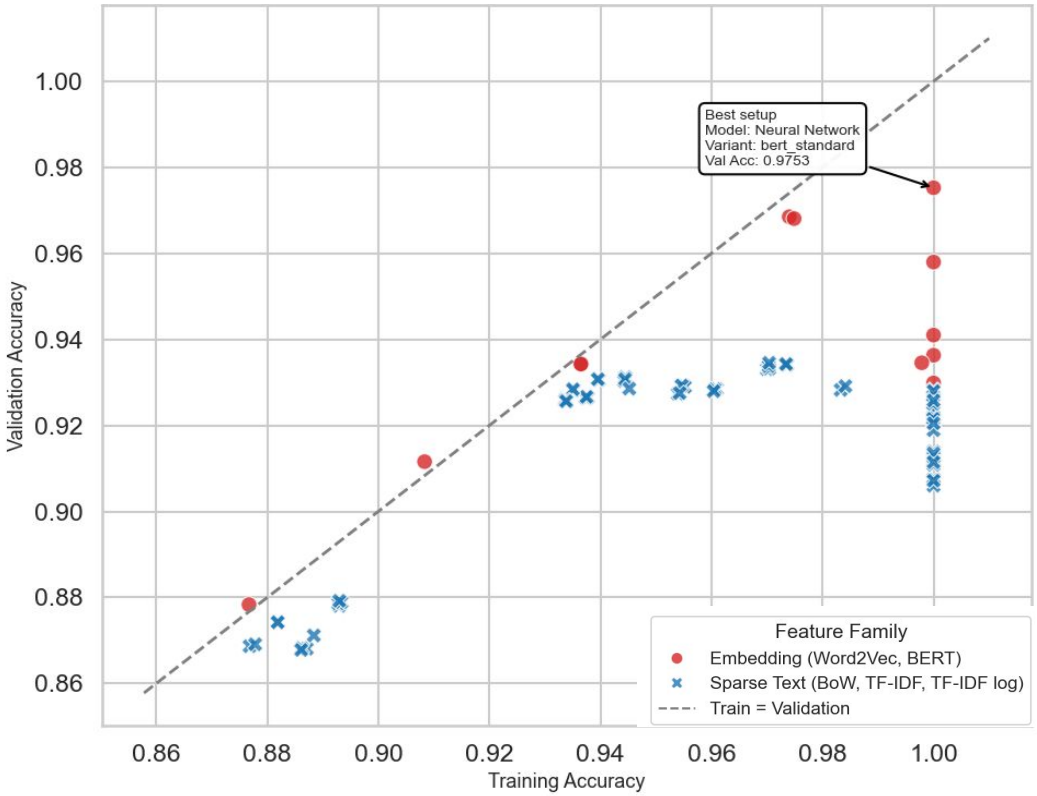
72 Sparse-Text Pipelines
4 (Preprocess) × 3 (Features) × 6 (Models) = 72

12 Embedding Pipelines
2 (Embeddings) × 6 (Models) = 12

Training vs Validation by Feature Family

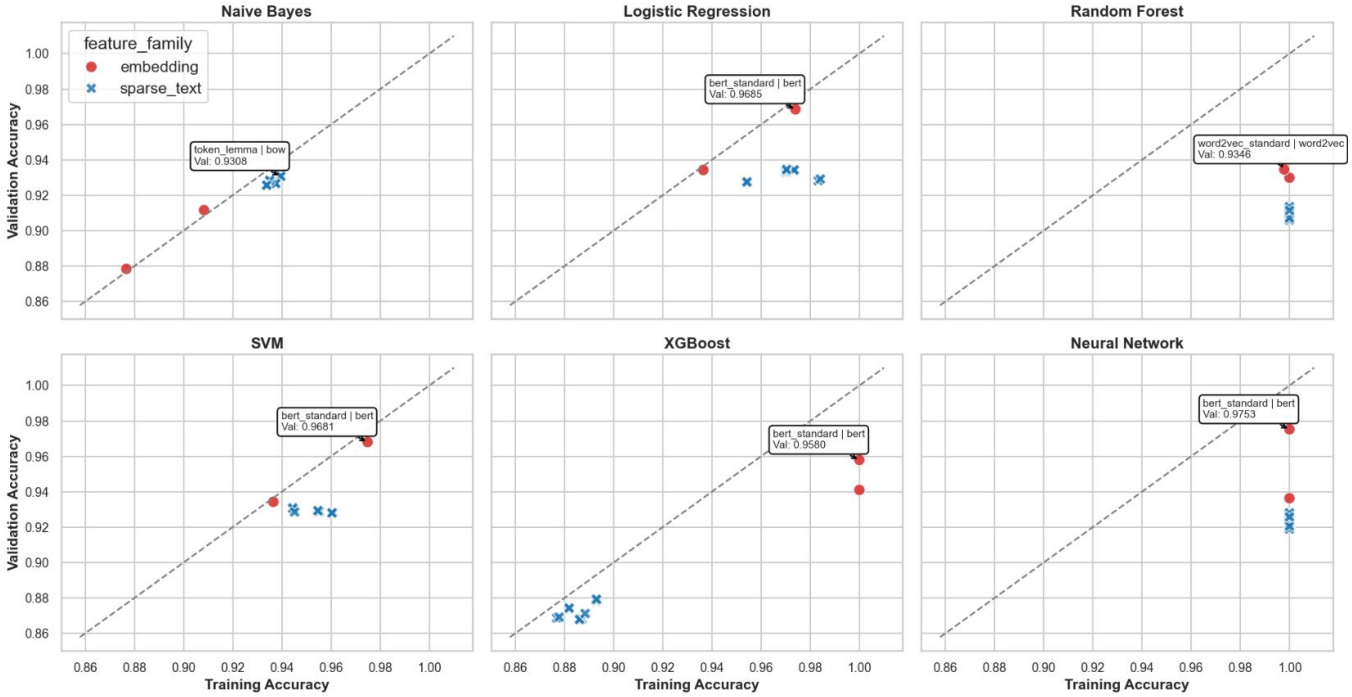
- Embedding-based methods, especially BERT, achieved the highest validation accuracy and included the best overall setup.

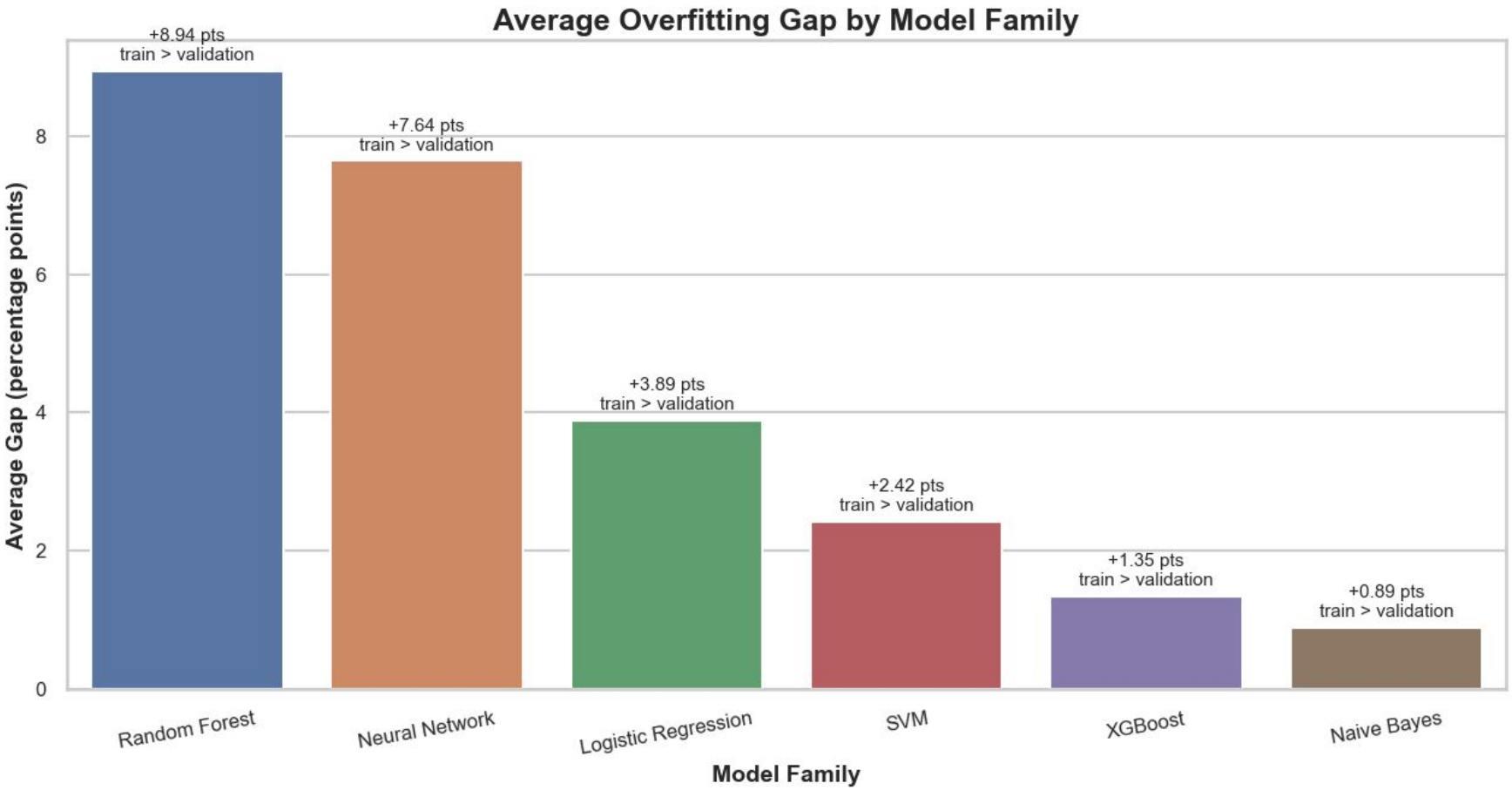
- Sparse-text methods were more tightly clustered, showing stable but generally lower performance than embeddings.



Training vs Validation: Feature and Model Families

- Embeddings helped Neural Network and XGBoost the most.
- Logistic Regression, SVM and Random Forest remained strong across both feature families.
- Naive Bayes benefited less.




Average Overfitting Gap by Model Family
+8.94 pts
train > validation
+7.64 pts
train > validation
+3.89 pts
train > validation
+2.42 pts
train > validation
+1.35 pts
train > validation
+0.89 pts
train > validation
Average Gap (percentage points)
0
2
4
6
8
Random Forest
Neural Network
Logistic Regression
SVM
XGBoost
Naive Bayes
Model Family

Recap:

- Feature representation mattered more than preprocessing.
- Sparse-text tweaks had little impact.
- Embedding-based methods, especially “BERT”, can outperform classical sparse-text approaches when paired with suitable models such as Neural Networks.
- Classical sparse-text pipelines, especially “TF-IDF + Logistic Regression”, remain strong and interpretable baselines.
- Complex models tend to overfit more, while simpler models generalize better.

Challenges:

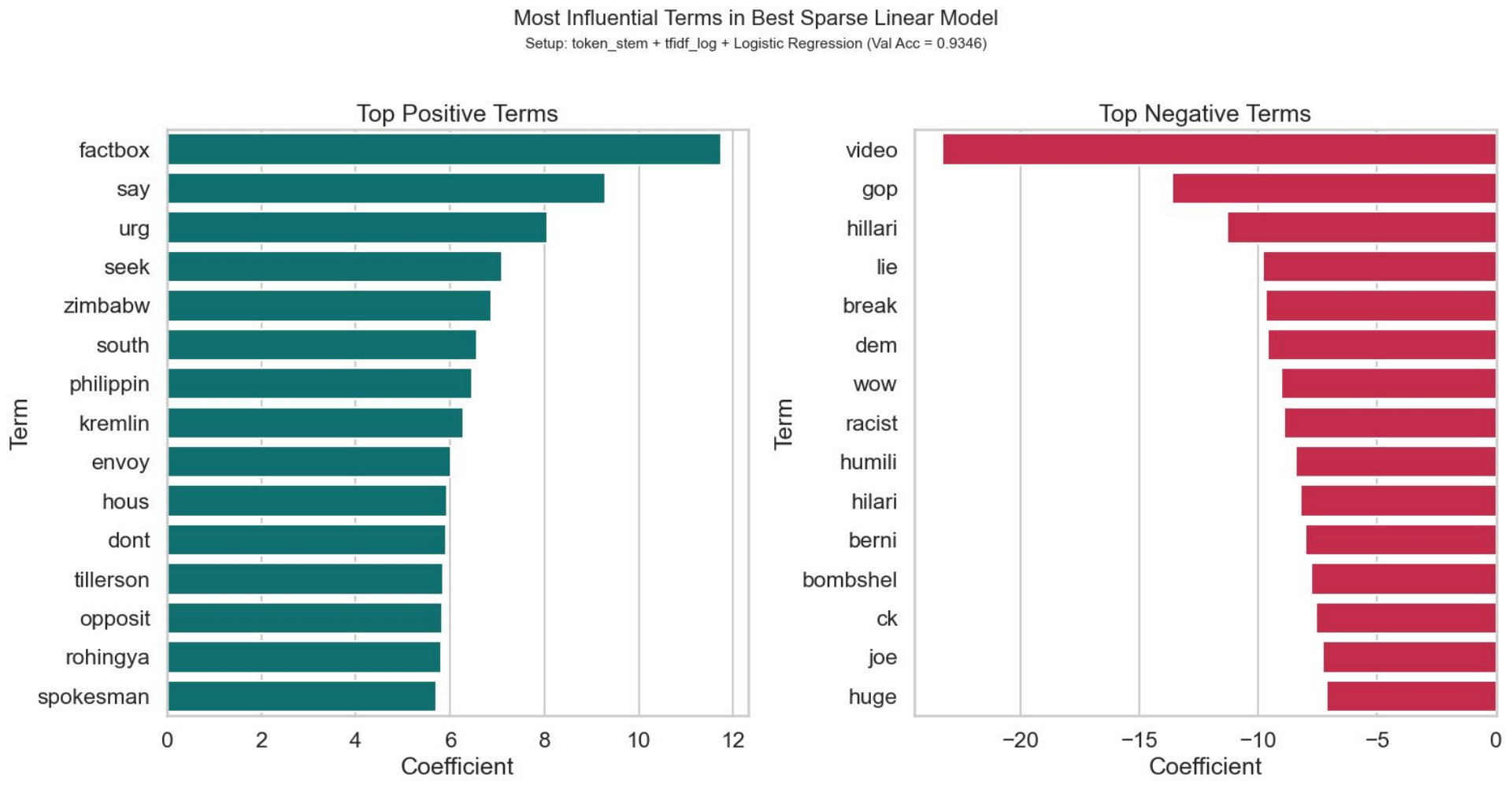
- 84 Pipelines meant long runtime.
- Used Pipeline and RandomizedSearchCV.

Final choice:

It depends on accuracy vs interpretability:

- Maximum performance → BERT + Neural Network
- Simplicity & efficiency → TF-IDF + Logistic Regression

Thank you for the attention


Most Influential Terms in Best Sparse Linear Model
Setup: token_stem + tfidf_log + Logistic Regression (Val Acc = 0.9346)
Top Positive Terms
factbox
say
urg
seek
zimbabw
south
philippin
kremlin
envoy
hous
dont
tillerson
opposit
rohingya
spokesman
Term
0
2
4
6
8
10
12
Coefficient
Top Negative Terms
video
gop
hillari
lie
break
dem
wow
racist
humili
hilari
berni
bombshel
ck
joe
huge
Term
−20
−15
−10
−5
0
Coefficient